

MATHIS JEGOU

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PROFILE

Mechatronics engineering student with 3 years of experience developing robotic and automated systems across mechanical design, embedded systems and prototyping. Seeking a U.S. engineering internship starting May 2027. Eligible for J-1 Internship Visa.

EDUCATION

Master's Degree in Mechatronics Engineering, ENSIBS Lorient *Expected Graduation: 2028*
Relevant Coursework: Embedded Systems, Programming, CAD, Sensors & Actuators, Robotics.

Technical Degree in Mechanical and Production Engineering, University of Rennes 2024 – 2025
Relevant Coursework: Mechanical Design, Finite Element Analysis, Materials Science, Prototyping

Pre-engineering undergraduate studies, Lycée Auguste Brizeux, France 2022 – 2024

Baccalauréat - Science (Math & Physics), Lycée Saint-Gabriel, France 2021

PROJECTS

Awake Challenge – French National Autonomous Vehicle Competition: Designed and built a 1:10 vehicle chassis, drivetrain and steering system in SolidWorks. Integrated LiDAR and sensors and developed the autonomous navigation system in ROS2 on a Raspberry Pi.

Autonomous Catamaran – GPS Waypoint Navigation: Designed a compact catamaran with a waterproof electronics compartment. Selected a Pixhawk flight controller for autonomous control and GPS waypoint navigation.

STM32 Mobile Robot: Implemented sensor-based obstacle detection, line following and closed-loop steering control on an STM32 platform using C.

4-Axis Robot Arm: Designed and sized a robot arm for a 5 kg payload at 0.4 m reach, targeting 90° rotation in 1s. Performed torque calculations, selected suitable motors and built the prototype using 3D printing.

Systems Engineering: Led a 5-person team to develop a carbon-neutral urban delivery system. Managed project planning, stakeholder requirements and performed functional analysis.

WORK EXPERIENCE

Mechanical Design Intern Jan 2025 – Mar 2025
Serma Inox - Designer and manufacturer of stainless steel process equipment *France*

- Reduced BOM preparation time by 90% by automating BOM extraction from SolidWorks and processing the extracted data in Excel using VBA.
- Designed mechanical assemblies in SolidWorks for food processing equipment.
- Created installation guides and user documentation for the automated BOM system, which improved usability for staff.

SKILLS

Technical Skills: CAD (SolidWorks, TopSolid), Programming (Python, C, VBA), Embedded Systems (Arduino, Raspberry Pi, STM32), Robotics (ROS2), Sensors (LiDAR), FEA (Abaqus), Prototyping (3D Printing)

Soft Skills: Leadership, Project Management, Problem Solving, Critical Thinking

Languages: English (Advanced), French (Native)